

market. In today's fast-paced corporate climate, where the capacity to develop and launch new services quickly can be a competitive edge, this dynamic scalability is essential. Furthermore, because cloud solutions eliminate the need for large capital investments in hardware and maintenance, they result in significant cost savings. So businesses optimise their IT costs by only paying for the resources they really utilise. Additionally, the cloud is perfect for international organisations because of its global reach, which enables smooth cross-border collaboration and data access. Essentially, the cloud offers the framework for contemporary, flexible, and economical corporate processes.

Cloud adoption: The rise of Big Data from 2013 to 2024

The necessity to manage massive amounts of data efficiently has led to an exponential growth in cloud use from 2013 to 2024. Businesses started looking into cloud solutions in the early 2010s, but the technology didn't really take off until the middle of the decade. Cloud service providers such as AWS, Microsoft Azure, and Google Cloud led the way in the mainstreaming of cloud computing by 2016. The cloud's capacity to provide scalable and affordable data processing and storage made it especially enticing to Big Data-related enterprises. Cloud platforms offered the infrastructure required to store, analyse, and extract insights from large datasets without requiring substantial on-premises resources as data volumes skyrocketed. Nearly 90% of businesses moved towards cloud services by 2020, using them for everything from machine learning to data analytics. The trend persisted into 2024, when enterprises were able to process petabytes of data in real-time thanks to cloud-native Big Data platforms like Snowflake and Databricks, which were essential to data strategy. This change has given

companies the ability to make data-driven decisions faster, which has fuelled innovation and given them a competitive edge in the digital era.

The terms of cloud computing and its offerings

The pay-as-you-go, on-demand distribution of IT resources over the internet is referred to as cloud computing. With this strategy, companies can concentrate on their core competencies instead of having to manage and operate physical servers and data centres. Infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS) are the three primary categories into which cloud services are generally divided (Table 1).

Infrastructure as a service (IaaS): IaaS provides virtualised computing resources over the internet. This includes virtual machines, storage, and networks. It is the most flexible cloud computing model, offering users complete control over their infrastructure. Companies use IaaS to avoid the cost and complexity of buying and managing their physical servers. Examples include Amazon Web Services (AWS) EC2, Microsoft Azure Virtual Machines, and Google Cloud Compute Engine.

Platform as a service (PaaS): PaaS offers a higher level of abstraction, providing not just infrastructure but also development tools, database management systems, and middleware. This allows developers to build, test, and deploy applications quickly without worrying about underlying hardware. PaaS is ideal for companies looking to streamline development workflows. Examples include Google App Engine, Microsoft Azure App Service, and Heroku.

Software as a service (SaaS): SaaS delivers fully functional applications over the internet, which users can access via a web browser. It eliminates the need for users

Table 1: The three primary categories of cloud services

Service model	Proposed year	Adoption ratio (2024)	Brief use case	Applied by	Implemented by	Proposed by
IaaS	Early 2000s	90%+	Full control over virtualised infrastructure	Enterprises, Cloud providers	Amazon Web Services, Microsoft Azure, Google Cloud	Amazon Web Services (AWS)
PaaS	Mid-2000s	70%+	Streamlining application development and deployment	Software developers, startups	Google App Engine, Microsoft Azure App Service, Heroku	Salesforce
SaaS	Late 1990s	95%+	Accessing software via the internet without installation	Businesses of all sizes, individuals	Salesforce, Google Workspace, Microsoft 365	Microsoft (Hotmail), Concur (initial SaaS concepts)